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Cloud Computing (CSA1513)

Experiment - 01

Cloud - Based Library Book Reservation System

Aim:-

To create a simple Cloud-based Library Book Reservation System for SIMATS Library using a cloud service provider and demonstrate SaaS.

Algorithm:-

1. Create a cloud-based library application.
2. Add book details such as Book ID, Title, Author, Publication, year, edition, No.of copies, Rack No, and Status.
3. Store the details in a cloud database.
4. Allow students to search and view available books.
5. Reserve an available book.
6. Update the status as Taken.
7. Change the status to Returned when the book is returned.

Cloud Service Provider

Firebase - Used for cloud database and hosting

Sample Input

BookID	Title	Author	Year	Copies	RackNO	status
B001	cloud comping	Sandhya	2023	5	R01	Returned
B002	AI	Rajitha	2022	3	R02	Taken

Output:-

SIMATS LIBRARY SYSTEM

Book :- Cloud Computing

Author :- Sandhya

Rack No:- R01

Status:- Returned

Reservation status :- Successful

Book status :- Taken

Result

The cloud-based Library Book Reservation System was successfully created, demonstrating the SaaS model where users access the application through the internet without installing it locally.

Experiment - 02

Cloud - Based Payroll Processing System

Aim:-

To create a simple cloud-based Payroll processing system using a cloud service Provider and demonstrate the SaaS Model.

Algorithm:-

1. Create a cloud-based Payroll application.
2. Enter employee details and salary information.
3. Calculate HRA, DA, TA, Gross Pay and Net Pay.
4. Store the payroll details in a cloud database
5. Display the employee salary details through the application

Cloud service Provider

Firebase - used for cloud database and hosting.

Required Fields

1. Employee Name
2. EMployee ID
3. Experience in Present Organization
4. Overall Experience
5. Basic Pay

6. HRA
7. DA
8. TA
9. Gross Pay
10. Net Pay

Sample Input:

Filed	Value
Employee Name	Sandhya
~~Employee ID~~	[illegible]
Present Org. Experience	2 years
Overall Experience	4 years
Basic Pay	₹ 30,000
HRA	₹ 6,000
DA	₹ 3,000
TA	₹ 2,000

GRoss Pay = Basic Pay + HRA + DA + TA = ₹ 41,000

Output:

Employee Name :- Sandhya
Basic Pay :- ₹ 30000
HRA :- ₹ 6000
DA :- ₹ 3000
TA :- ₹ 2000
Gross Pay :- ₹ 41,000
Net Pay :- ₹ 41,000

Payroll Generated successfully

Result:-
The Cloud-Based Payroll Processing System was successfully created using a cloud service, demonstrating SaaS, where employees/admins can access payroll servies through the internet.

Experiment-03

Virtualization

Aim:-
To demonstrate virtualization by installing a Type-2 Hypervisor and creating a Virtual Machine with 1 CPU, 2GB RAM and 15GB Storage.

Software Required:-

* Oracle Virtual Box - Type-2 Hypervisor
* Host OS - Windows/Linux
* Guest OS - Ubuntu/Linux

Procedure:-

1. Install Oracle Virtual Box on the host operating System.
2. Open Virtual Box and Select New to Create a VM.
3. Enter the VM name and select the guest OS.
4. Allocate 1 CPU
5. Allocate 2GB RAM
6. Create a virtual hard disk of 15GB
7. Attach the guest OS ISO image.
8. Start the VM and install the guest OS.
9. Verify the VM configuration and boot the System.

Configuration

Resource	Configuration
Hypervisor	Oracle VirtualBox
Type	Type-2
CPU	1 Core
RAM	2 GB
Storage	15GB
Host OS	Windows/Linux
Guest OS	Ubuntu/Linux

Result:-

A Virtual Machine was successfully created using the Type-2 Hypervisor Oracle Virtual Box with 1 CPU, 2 GB RAM and 15 GB Storage, demonstrating virtualization.

Experiment - 04

VM Cloning Using Type-2 Hypervisor

Aim:-

To demostrate virtualization by installing a Type-2 Hypervisor, Creating a Virtual Machine, cloning it and testing the cloned VM.

Software Required:-

* Oracle virtual Box
* Host OS - Windows/Linux
* Guest OS - Ubutu/Linux

Procedure:-

1. Install and open Oracle VirtualBox and Create configure a virtual machine.
2. Install the required Guest OS in the VM and shut down the VM.
3. Right click the VM and select clone.
4. Enter a name for the cloned VM.
5. Select Full clone and create the clone and start the cloned VM.
6. Verify that the cloned VM contains the same OS and settings as the original VM.
7. Test the cloned VM by loading and running the previous version.

Configuration:-

Parameter	Value
Hypervisor	Oracle Virtual Box
Type	Type-2
Original VM	Ubuntu
Clone VM	Ubuntu-clone
CPU	1 core
RAM	2GB
Storage	15GB

Result:-

The virtual Machine was successfully cloned using Oracle virtual Box. The cloned VM was loaded and tested successfully, demonstrating VM cloning and virtualization.